Code-Layout, Readability and Reusability

Date	02 October 2026
Team ID	SWTID-2026-1529
Project Name	EduGenie: Google gemini powered learning assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughou	the code	
2	Proper File Structure	Files and folders are logically organize	d	
3	Meaningful Variable Names	Variables reflect their purpose clearly		
4	Function / Method Names	Functions are descriptively named		
5	Code Comments	Inline and block comments explain log	c	
6	Modular Design	Code is split into reusable functions/m	dules	
7	No Redundant Code	Duplicate or unused code is removed		
8	Error Handling	Exceptions and errors are handled gra	cefully	

The codebase uses a decoupled architecture with a FastAPI backend (Python) and a Next.js 15 frontend.
All backend logic is modularized into dedicated modules (e.g., qna.py, quiz_module.py).
State management on the React side uses modular hooks, while the UI is built with reusable glassmorphism components.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	JSON Validator (utils.py)	Strips markdown and validates schema		Ensures app never crashes from AI c
1 2	Glassmorphic UI Engine	React UI components for interactive state		Reusable across all Next.js pages

2				
3				
4				
5				

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Strict separation of concerns between Next.js and FastAPI
Readability	5	Clear variable names and comprehensive docstrings in Python
Reusability	5	Modules like utils.py and UI components are highly reusable
Documentation / Comments	5	Centralized error handling in FastAPI main router
Overall Score		All Gemini prompt logic is isolated and version-controlled

Overall Score: 25/25